Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State the negligible-air-resistance assumption.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your model or result. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Bring back an earlier idea

Outside a spherical source, field at r is 75 N/kg.
Find field at 5 r and the force on 0.7 kg there.

Word help: field = a description of an influence at each location

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_x = 0$; $a_y = -g$; v_x = constant.
Upward positive; g = 10 m/s². Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at 3 m/s and falls for 2 s. Horizontal travel is 3×2=6 m. With g=10 m/s² and no initial vertical velocity, its final vertical velocity is 20 m/s down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after 2 s.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

One ball is dropped and another launched horizontally from the same height at the same instant.
Compare fall times and explain.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Solve and justify

A ball launches horizontally at 8 m/s and falls for 1.5 s. Find$\Delta x, \Delta y$ and final velocity components using upward positive.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student says a larger horizontal speed holds the ball up longer.
Explain which force or acceleration would be needed for that claim.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

For $x_i=4\pm1$ m, $v=6\pm0.5$ m/s and exact $t=8$ s, find the central final position and conservative interval.
Name a specific check of your method.

Word help: conservative bound = a limit chosen to include every allowed case

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A projectile's velocity is(6,-8) m/s.
Find speed. What information is still needed to determine its position?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A projectile has initial velocity (6, 8) m/s; use upward positive and $g=10$ m/s².
Find velocity after 1 s and explain whether its horizontal motion depends on its vertical velocity.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3